Worksheet 4-5 D Related Rates #4
AP Calculus Name_____

Directions: Solve the following problems. Show all work clearly and neatly ON NOTEBOOK PAPER.

1. If xy = −3 and $\frac{dx}{dt} = 1$, then find $\frac{dy}{dt}$ when x = 6.

2. The volume of a cube is decreasing at the rate of 10 m^3/hr. How fast is the total surface area decreasing when the surface area is 54 m^2?

3. Rachael is blowing up a balloon so that the diameter increases at the rate of 10 cm/sec. At what rate must she blow air into the balloon when the diameter measures 4 cm? Volume of a sphere is $V = \frac{4}{3}\pi r^3$.

4. Assume Clark and Lana leave Smallville Stadium from the same point at the same time. If Clark runs south at 4 mph and Lana runs west at 3 mph, how fast will the distance between Clark and Lana be changing at 10 hours?

5. Suppose Aaron is pumping water into a tank (in the shape of an inverted right circular cone) at a rate of 1600 ft^3/min. If the altitude is 10 ft and the radius of the base is 5 ft, find the rate at which the radius is changing when the height of the water is 7 ft. Volume of a Cone: $V = \frac{1}{3}\pi r^2 h$.

6. The talented Ed Wynwyte is flying a kite at a constant height of 400 meters. The kite is moving horizontally at a rate of 30 m/sec. How fast must he unwind the string when 500 m of string is paid out?

7. A ladder 15 feet tall leans against a vertical wall of a home. If the bottom of the ladder is pulled away horizontally from the house at 4 ft/sec, how fast is the top of the ladder sliding down the wall when the bottom of the ladder is 9 feet from the wall?

8. A cone (vertex down) with height 10 inches and radius 2 inches is being filled with water at a constant rate of 2 in^3/sec. How fast is the surface of the water rising when the depth of the water is 6 inches?

9. A streetlight is 15 feet above the sidewalk. Jonathan, who is 7 feet tall, walks away from the light at the rate of 5 feet per second. Determine the rate at which Jonathan's shadow is lengthening at the moment that he is 20 feet from the base of the light.

10. A spherical balloon is inflated with helium at the rate of 100π ft^3/min. How fast is the balloon's radius increasing at the instant the radius is 5 ft? How fast is the surface area increasing?
Volume of a sphere is $V = \frac{4}{3}\pi r^3$ Surface Area of a sphere is $A = 4\pi r^2$

11. Two parallel sides of a rectangle are being lengthened at the rate of 2 in/sec., while the other two sides are shortened in such a way that the figure remains a rectangle with constant area A of 50 square inches. What is the rate of change of the perimeter P when the length of an increasing side is 5 in?

12. The number of fast food restaurants in a town can be modeled by the function F defined by $F(x) = \frac{40x}{x+2}$ where x is the number of Uber drivers in the town. When there are 24 fast food restaurants in the town, the number of Uber drivers in the town is increasing at a rate of $\frac{1}{8}$ drivers per month. What is the rate of change of the number of fast food restaurants in the town, in restaurants per month, when there are 24 fast food restaurants in the town?

Answers:

1.) $\frac{1}{12}$ 2.) The total surface area is decreasing at a rate of $\frac{40}{3}$ m^2/h 3.) She must blow air into the balloon at a rate of 80π cm^3/sec

4.) The distance between Clark and Lana is increasing at a rate of 5 mi/h at 10 hours 5.) The radius is increasing at a rate of $\frac{3200}{49\pi}$ ft/min when the height is 7 ft

6.) He must unwind the string at a rate of 18 m/sec 7.) The top of the ladder is sliding down the wall at a rate of 3 ft/sec

8.) The surface of the water is rising at a rate of $\frac{25}{18\pi}$ in/sec 9.) His shadow is lengthening at a rate of $\frac{35}{8}$ ft/sec

10.) The balloon's radius is increasing at a rate of 1 ft/min and the surface area is increasing at a rate of 40π ft^2/min

11.) The perimeter is decreasing at a rate of 4 in/s when the length of an increasing side is 5 in.

12.) The number of fast food restaurants in town is increasing at a rate of $\frac{2}{5}$ restaurants/month when there are 24 restaurants